

INTELLIGENT WATER BODY MAPPING FROM MULTI-SOURCE SATELLITE OBSERVATIONS USING DEEP LEARNING

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Course: Essentials of Ocean Science

Instructor: Prof. Dr. Oliver Zielinski



ABOUT THE PROJECT

Accurate water-body detection is essential for environmental monitoring, flood assessment, coastal management, and climate-change studies. However, identifying water from satellite imagery can be challenging due to clouds, shadows, seasonal variations, and the complexity of different landscapes.

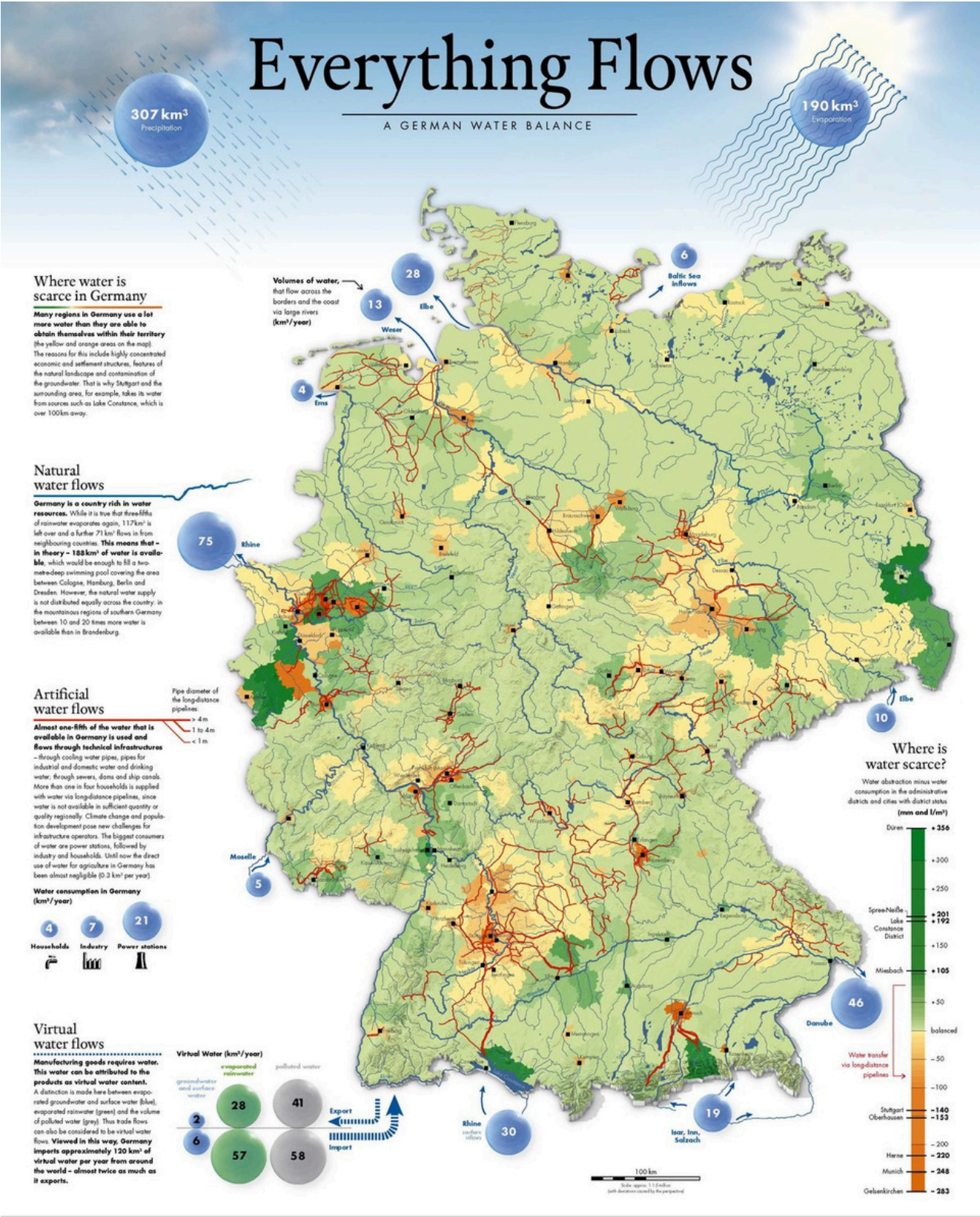
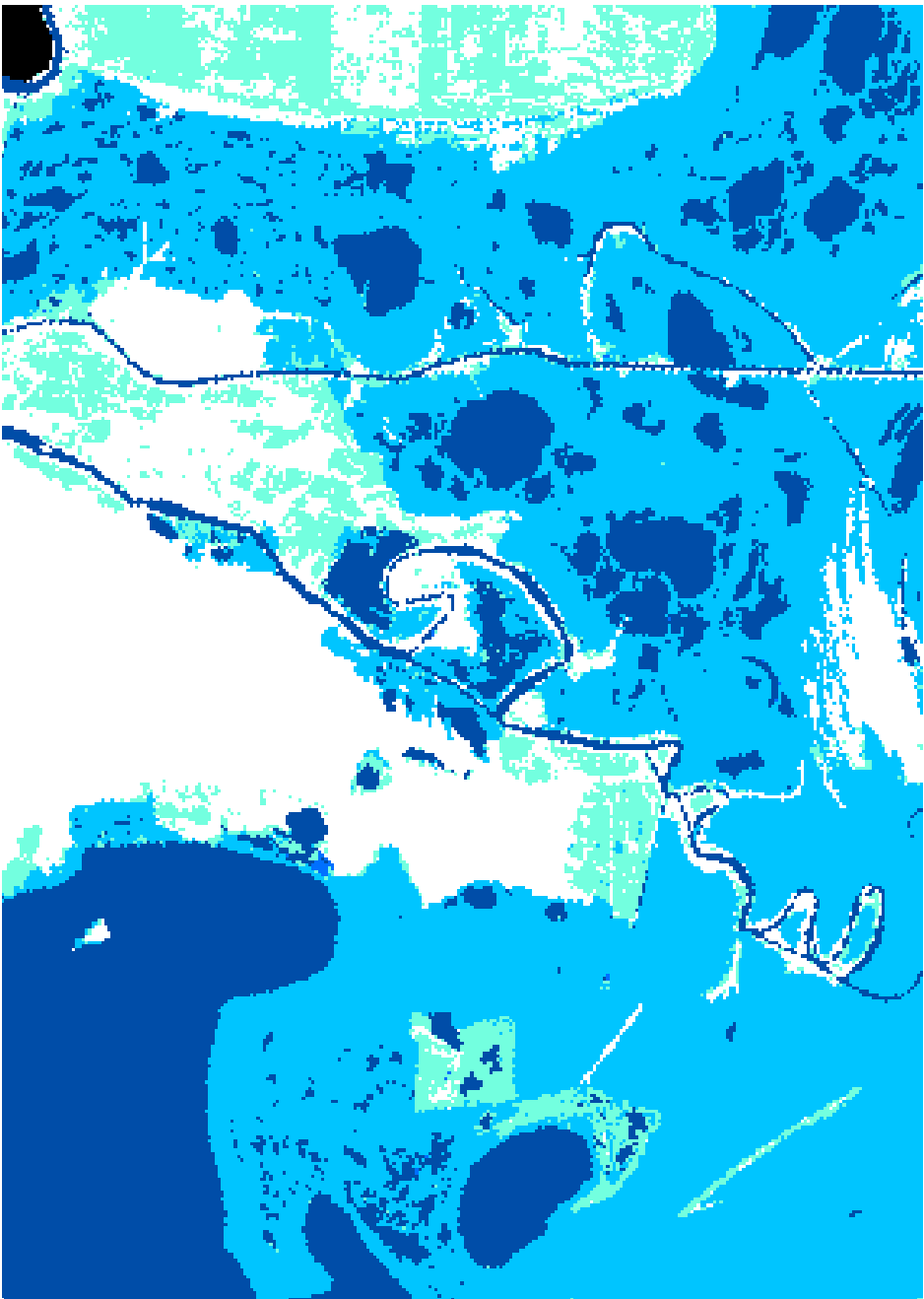
To address this challenge, this project uses a U-Net deep learning model trained on 12-channel multi-source satellite data. The input data combines spectral information from Sentinel-2/Landsat imagery with elevation models, land-cover maps, quality assessment layers, and historical water occurrence information.

By integrating these complementary data sources, the model can better distinguish water from non-water regions and automatically generate water masks, providing a scalable solution for water-resource monitoring and ocean science applications.

WHY IS WATER MAPPING IMPORTANT?

Water bodies continuously change due to:

- Climate change
- Flooding
- Coastal erosion
- Sea level rise
- Human activities



"Water is one of the most dynamic components of the Earth system. Monitoring its spatial distribution is essential for understanding environmental change, coastal processes, and aquatic ecosystems."



Develop an AI model capable of:

- Detecting water bodies
- Producing accurate water masks
- Supporting ocean and environmental monitoring

Can a U-Net deep learning model accurately segment water bodies using multi-source satellite imagery and geospatial datasets?



PROJECT

OBJECTIVE



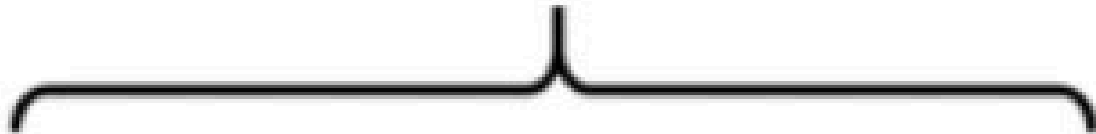
Connection to Ocean Science

This project can support:

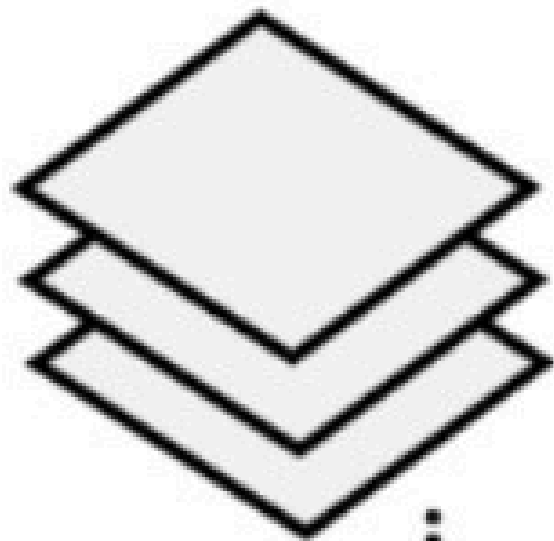
- Coastal monitoring
- Estuary mapping
- Wetland assessment
- Flood detection
- Tracking shoreline changes
- Climate impact studies

OCEAN SCIENCE RELEVANCE





Harmonized Sentinel-2/Landsat



Coastal aerosol

Blue

Green

Red

NIR

SWIR1



SWIR2

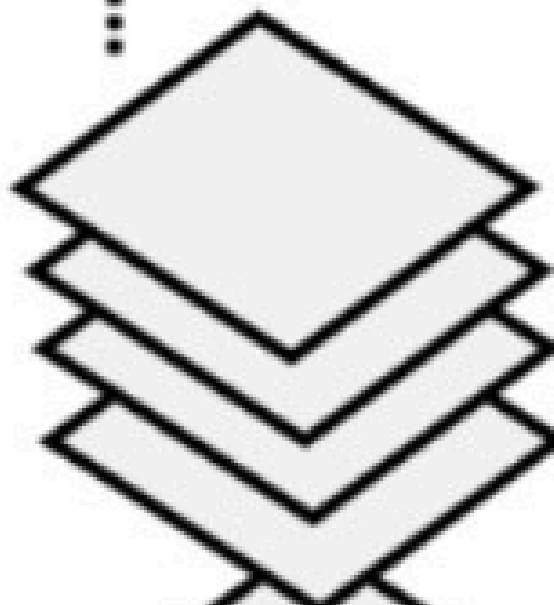
QA Band

Merit DEM

Copernicus DEM

ESA world cover map

Water occurrence probability



Binary water mask
(separate file)



Ground sampling distance = 30m
Patch size = 512 pixels

DATASET OVERVIEW



The model uses:

Band	Purpose	Layer	Purpose
Coastal Aerosol	Coastal water observation	SWIR2	Surface characterization
Blue	Water penetration	QA Band	Cloud quality control
Green	Vegetation-water contrast	MERIT DEM	Elevation information
Red	Surface reflectance	Copernicus DEM	Terrain information
NIR	Water strongly absorbs NIR	ESA World Cover	Land cover classification
SWIR1	Moisture detection	Water Occurrence Probability	Historical water presence

RGB Image (Channels 1, 2, 3)



Channel 0



Channel 1



Channel 2



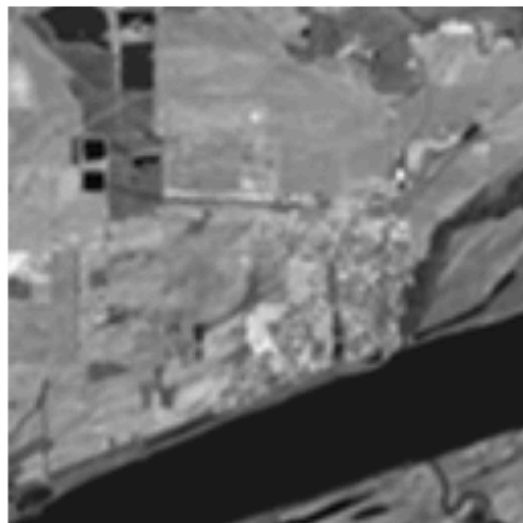
Channel 11



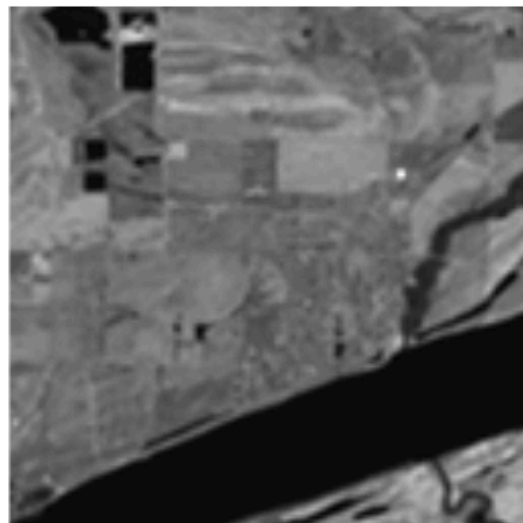
Channel 3



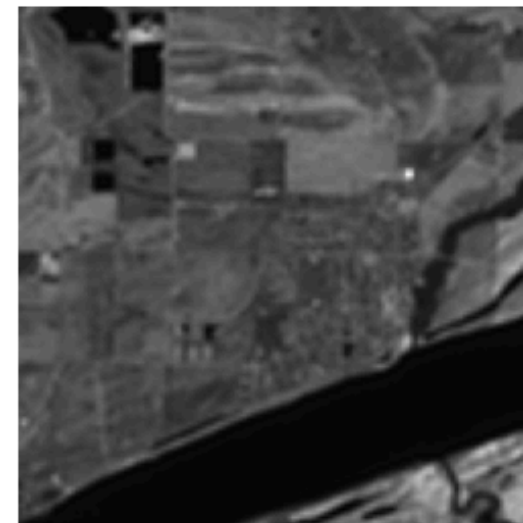
Channel 4



Channel 5



Channel 6



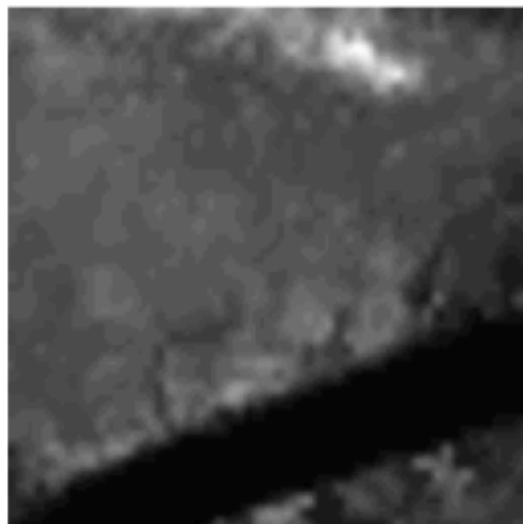
Label



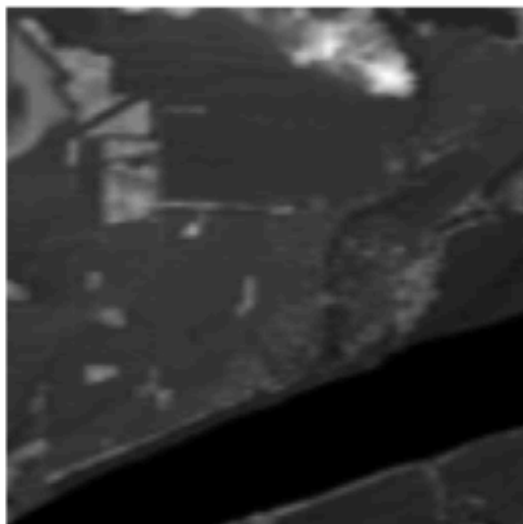
Channel 7



Channel 8



Channel 9



Channel 10



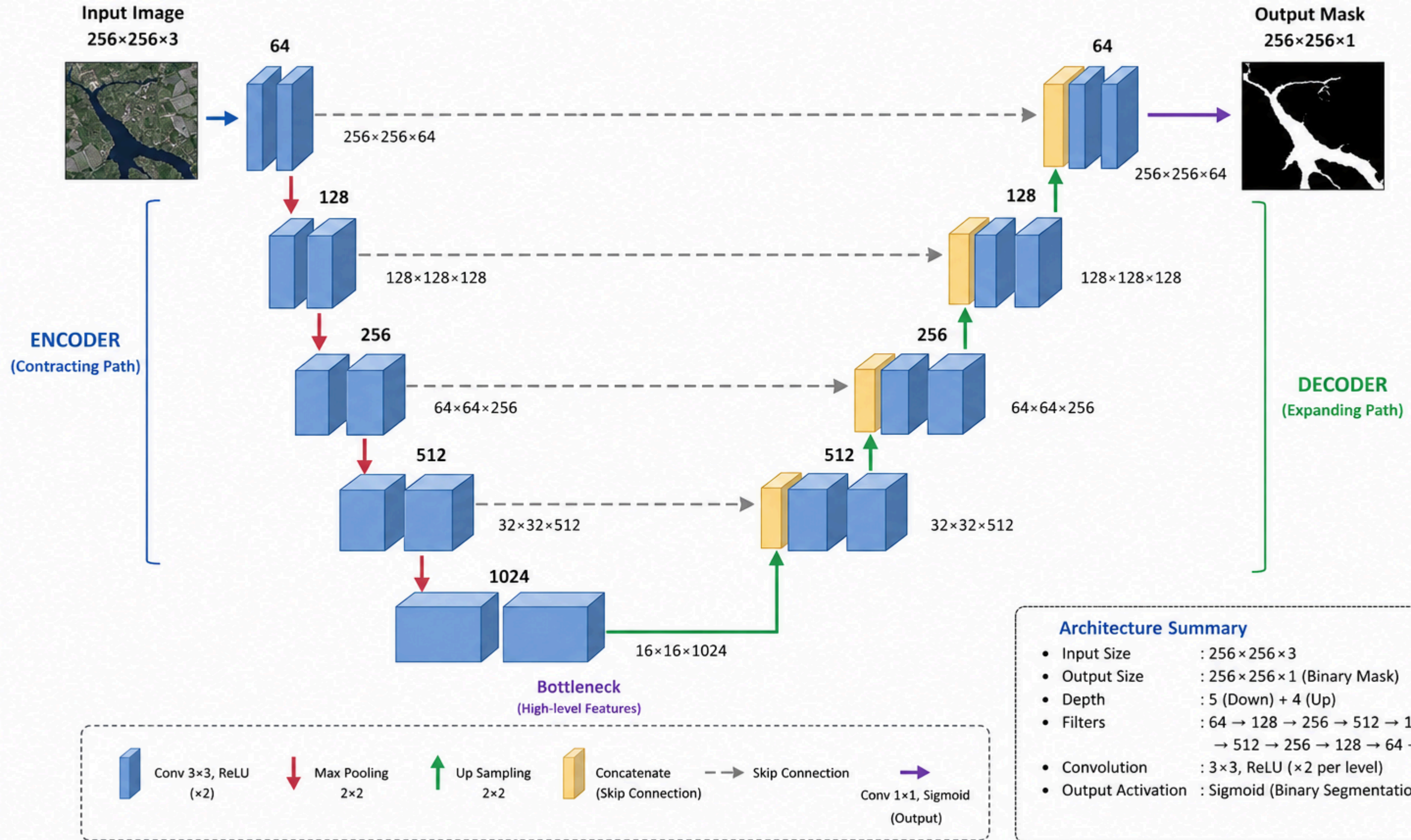
Patch Size: 512×512 pixels Ground Sampling Distance: 30 m Meaning:
Each pixel represents 30 meters on the Earth's surface.

Limited dataset size
(305 samples)

Data < [Link](#) >

WHY U-NET?

U-NET ARCHITECTURE



U-Net is a fully convolutional network with symmetric encoder-decoder structure and skip connections that help preserve spatial details for precise segmentation.

U-NET ARCHITECTURE

Originally developed for image segmentation.

Very effective because:

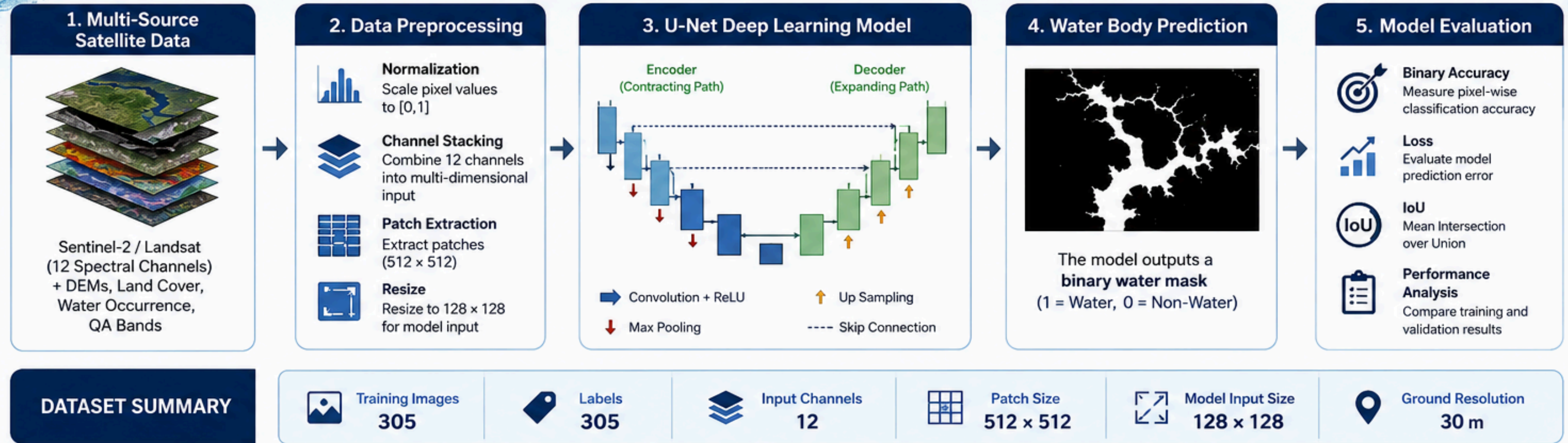
- Learns spatial patterns
- Preserves fine boundaries
- Works well with limited training data
- Produces pixel-level classification

U-Net is one of the most widely used architectures for segmentation tasks.

Input Satellite Image → Predicted Water Mask

METHODOLOGY

AI-Based Water Body Segmentation Workflow



RESULTS & PERFORMANCE EVALUATION

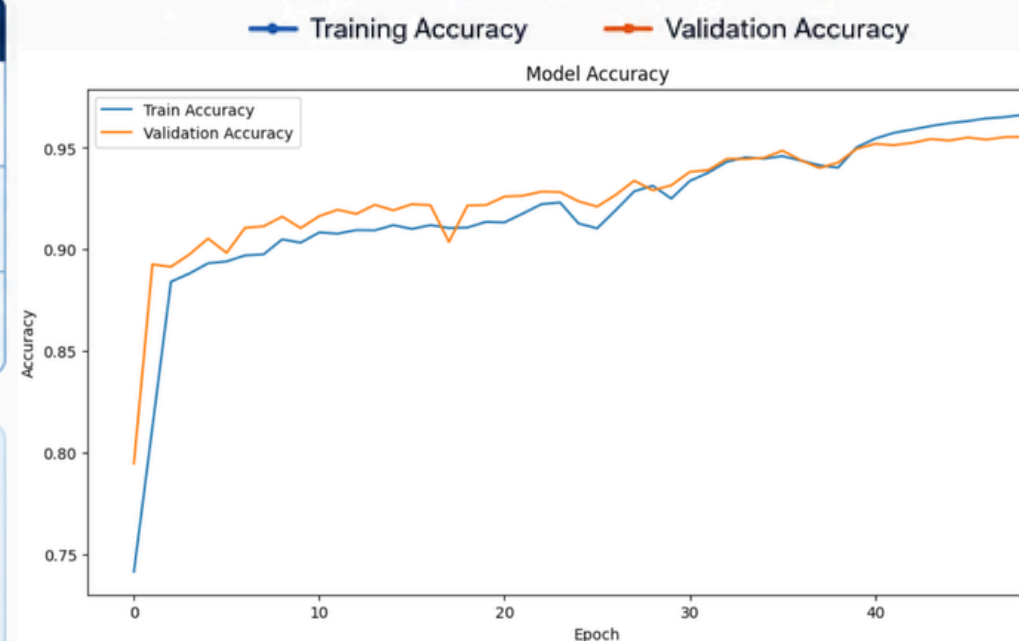
Model Performance

Metric	Training	Validation
Loss	0.0785	0.1425
Binary Accuracy	96.76%	95.63%
Mean IoU	0.4162	0.4118

Performance Interpretation

The model achieved strong classification performance with validation accuracy exceeding 95%, demonstrating its ability to distinguish water from non-water regions. The consistent Mean IoU values between training and validation datasets indicate stable segmentation behavior across unseen samples.

Training & Validation Accuracy



Final Training Accuracy
96.76%

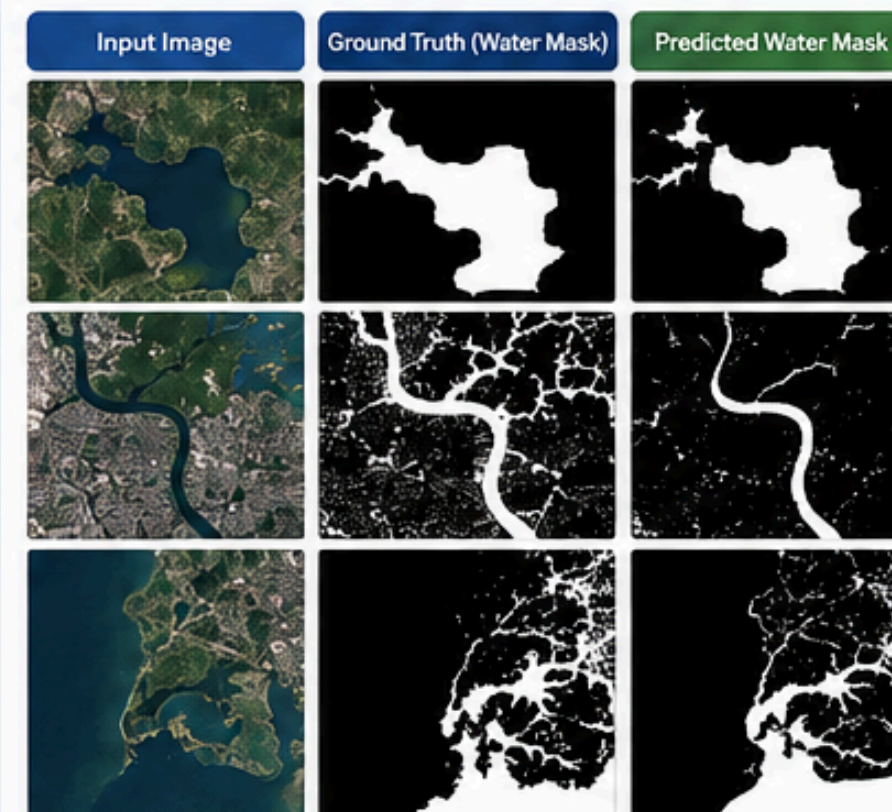


Final Validation Accuracy
95.63%



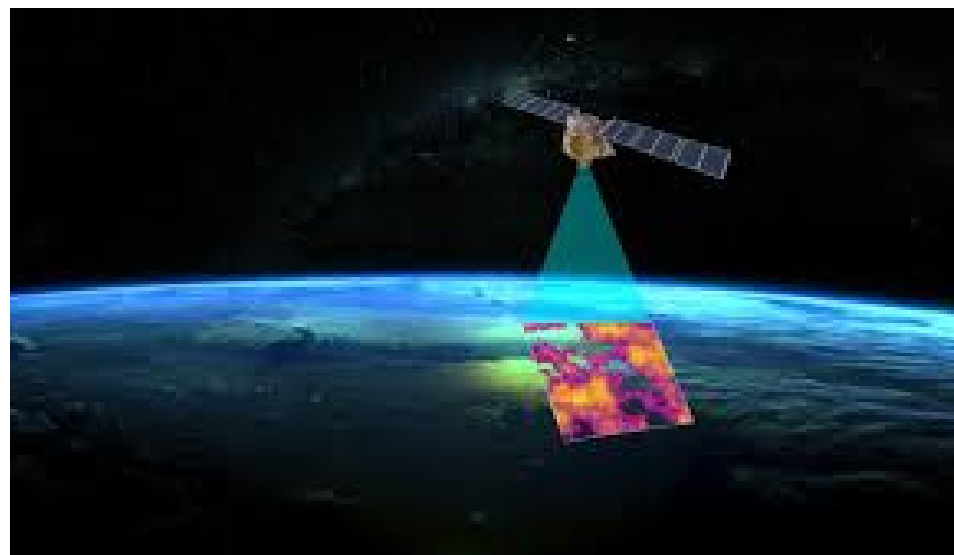
Final Mean IoU
0.4118

Example Segmentation Results



Key Findings

- ✓ Validation Accuracy reached 95.63%
- ✓ High training accuracy of 96.76%
- ✓ Consistent performance with small gap between training and validation
- ✓ Mean IoU above 0.41 indicates good overlap in water segmentation
- ✓ Suitable for large-scale environmental monitoring and water mapping applications



Why This Approach Is Powerful

- ✓ Fully automated
- ✓ Scalable globally
- ✓ Works on large satellite datasets
- ✓ Reduces manual mapping
- ✓ Useful for near real-time monitoring



ADVANTAGES



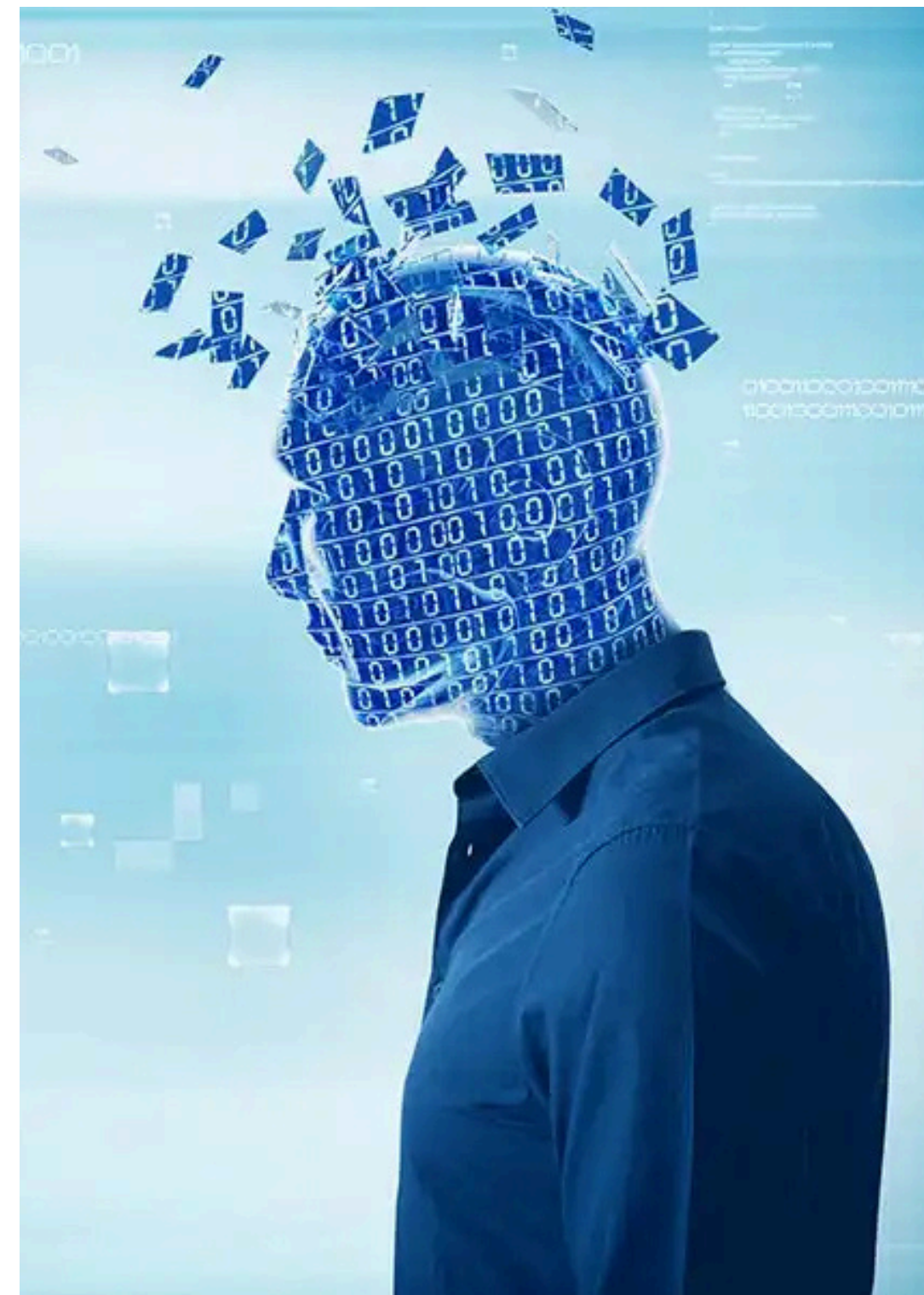
Challenges

- Clouds
- Shadows
- Mixed pixels
- Optimized for 12-channel satellite data
- Seasonal variation
- Small water bodies
- Limited dataset size
- Reduced accuracy on RGB-only images

Potential improvements:

- Attention U-Net
- DeepLabV3+
- Transformer-based segmentation
- Dedicated RGB-compatible version

LIMITATIONS



TRY MY MODEL :)

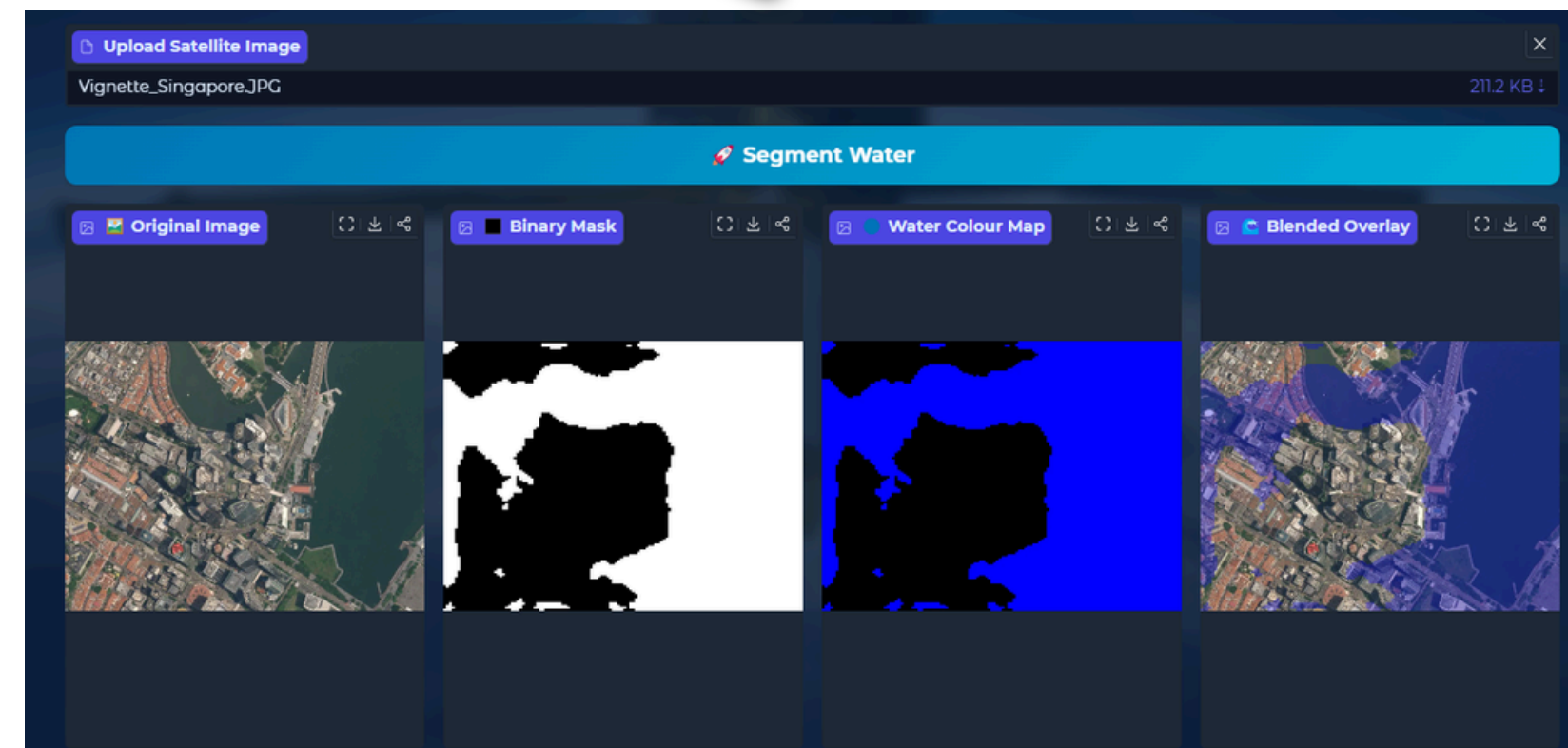
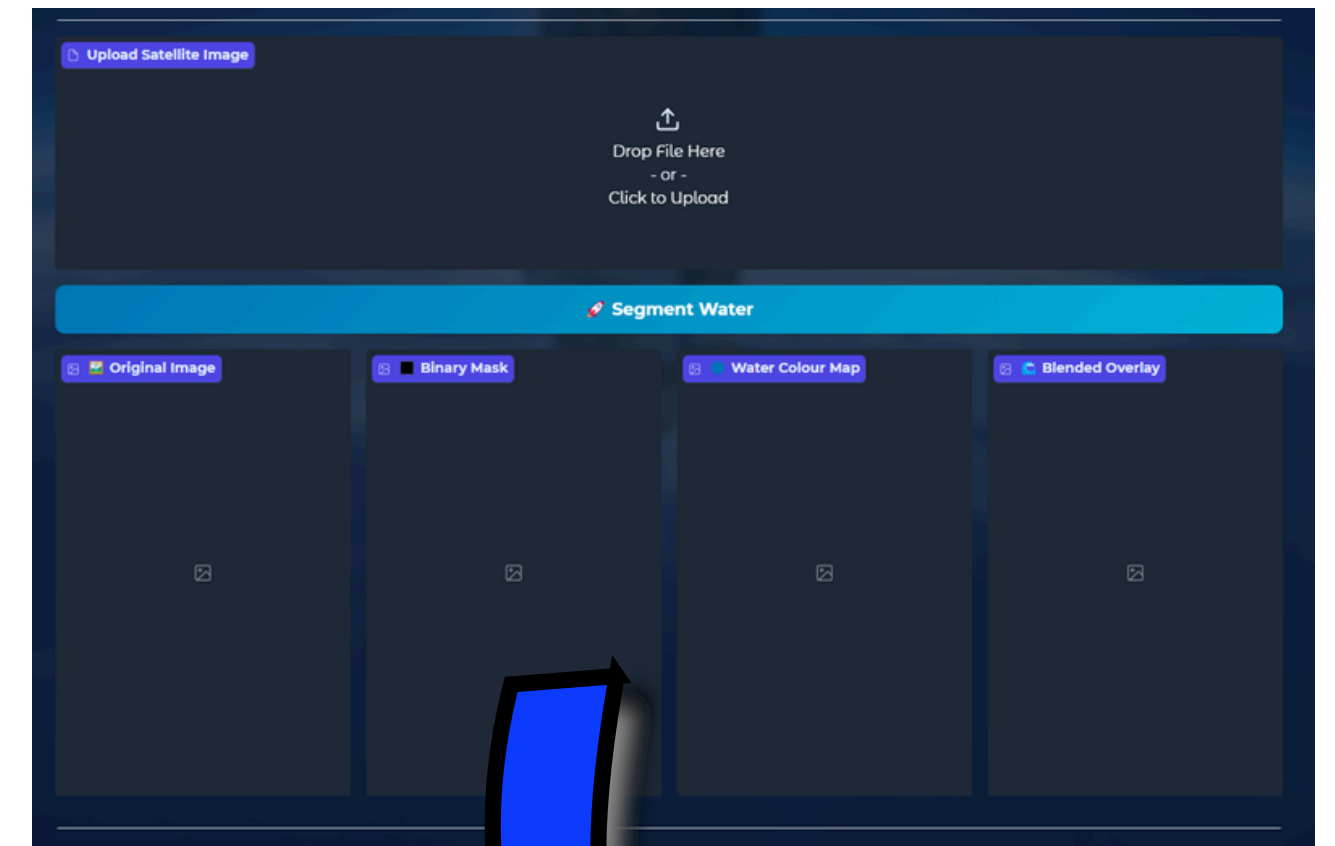
1-Select Satalite 12-channels Image :
(Here)

2-go through this link and upload the Image :
(Live Demo)



Und... läuft!

Repo Link





ABOUT ME

Master's student at rostock universitat specializing in Ocean Engineering and Renewable Energy Technologies, with a foundation in Mechatronics and Ai . Interested in renewable energy technologies, wind energy, digitalization, and AI-driven solutions that accelerate the transition to sustainable energy.



DATA SCIENTIST



COMPUTER VISION



NLP ENGINEER



AI ENGINEER



INTERSHIPS

Thank you for your time.

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